Determination of Refractive Index and Molecular Refractivity

Aim

To determine the refractive indices of the given liquids: water, ethanol, glycerol, and cyclohexane using an Abbe refractometer, and to calculate their molecular refractivities.

Principle

The refractive index of a liquid is the ratio of the velocity of light in vacuum to its velocity in the liquid:

$$n = \frac{c}{v}$$

An Abbe refractometer determines the refractive index from the critical angle at the interface between the sample and a high-refractive-index prism. The boundary between the illuminated and dark regions in the telescope is adjusted to the cross-wire, and the refractive index is read directly from the instrument.

The molecular refractivity is calculated using the Lorentz–Lorenz equation:

$$R_m = \left(\frac{n^2 - 1}{n^2 + 2}\right) \frac{M}{\rho}$$

where:

- R_m = molecular refractivity, in $\mathrm{cm^3\,mol^{-1}}$
- n = refractive index of the liquid
- M = molar mass of the liquid, in $\mathrm{g\,mol^{-1}}$
- ρ = density of the liquid, in $\mathrm{g\,cm^{-3}}$

Thus, once n, M, and ρ are known, molecular refractivity can be calculated.

Apparatus and Chemicals Required

Apparatus

- Abbe refractometer
- Dropper

- Soft lens tissue or lint-free tissue
- Distilled water

Chemicals

- Water
- Ethanol
- Glycerol
- Cyclohexane

Procedure

1. Clean the prism surfaces of the Abbe refractometer carefully using a suitable solvent and soft lens tissue.
2. Place a few drops of the liquid sample on the lower prism so that the entire prism surface is covered.
3. Close the upper prism gently to form a thin, uniform layer of the liquid between the prisms.
4. Direct the light source toward the refractometer.
5. Look through the eyepiece and adjust the illumination so that a distinct light–dark boundary is visible.
6. Adjust the dispersion compensator, if provided, to remove colour fringes and obtain a sharp boundary.
7. Rotate the measuring control until the light–dark boundary coincides exactly with the cross-wire.
8. Read the refractive index from the refractometer scale.
9. Record the refractive index of the liquid.
10. Clean the prisms thoroughly before measuring the next liquid.
11. Repeat the procedure for water, ethanol, glycerol, and cyclohexane.
12. Using the measured refractive index and the density and molar mass of each liquid, calculate molecular refractivity using the Lorentz–Lorenz equation.

Table 1: Refractive indices and molecular refractivities of the given liquids

Liquid	M $\mathrm{g\,mol^{-1}}$	ρ $\mathrm{g\,cm^{-3}}$	n RI	R_m $\mathrm{cm^3\,mol^{-1}}$
Water				
Ethanol				
Glycerol				
Cyclohexane				

Observation Table

Calculations

For Water

Given:

$$n = 1.334, \qquad M = 18\ \mathrm{g\,mol^{-1}}, \qquad \rho = 0.998\ \mathrm{g\,cm^{-3}}$$

Using the Lorentz–Lorenz equation,

$$R_m = \left(\frac{n^2 - 1}{n^2 + 2}\right)\frac{M}{\rho}$$

$$R_m = \left(\frac{(1.334)^2 - 1}{(1.334)^2 + 2}\right) \times \frac{18}{0.998}$$

$$R_m \approx 3.720\ \mathrm{cm^3\,mol^{-1}}$$

Result

The refractive indices of the given liquids are determined using an Abbe refractometer, and their molecular refractivities are calculated using the Lorentz–Lorenz equation.

Precautions

1. The prism surfaces must be completely clean and dry before every measurement.
2. Only a few drops of sample should be used; the entire prism surface should be uniformly covered.
3. Avoid air bubbles between the prisms.
4. The temperature should be kept constant because refractive index is temperature-dependent.
5. The light–dark boundary should be sharply focused before taking the reading.
6. The cross-wire should be positioned exactly at the boundary.

7. Volatile liquids such as acetone and methanol should be measured promptly after placing them on the prism.

8. Do not scratch or damage the prism surfaces while cleaning them.